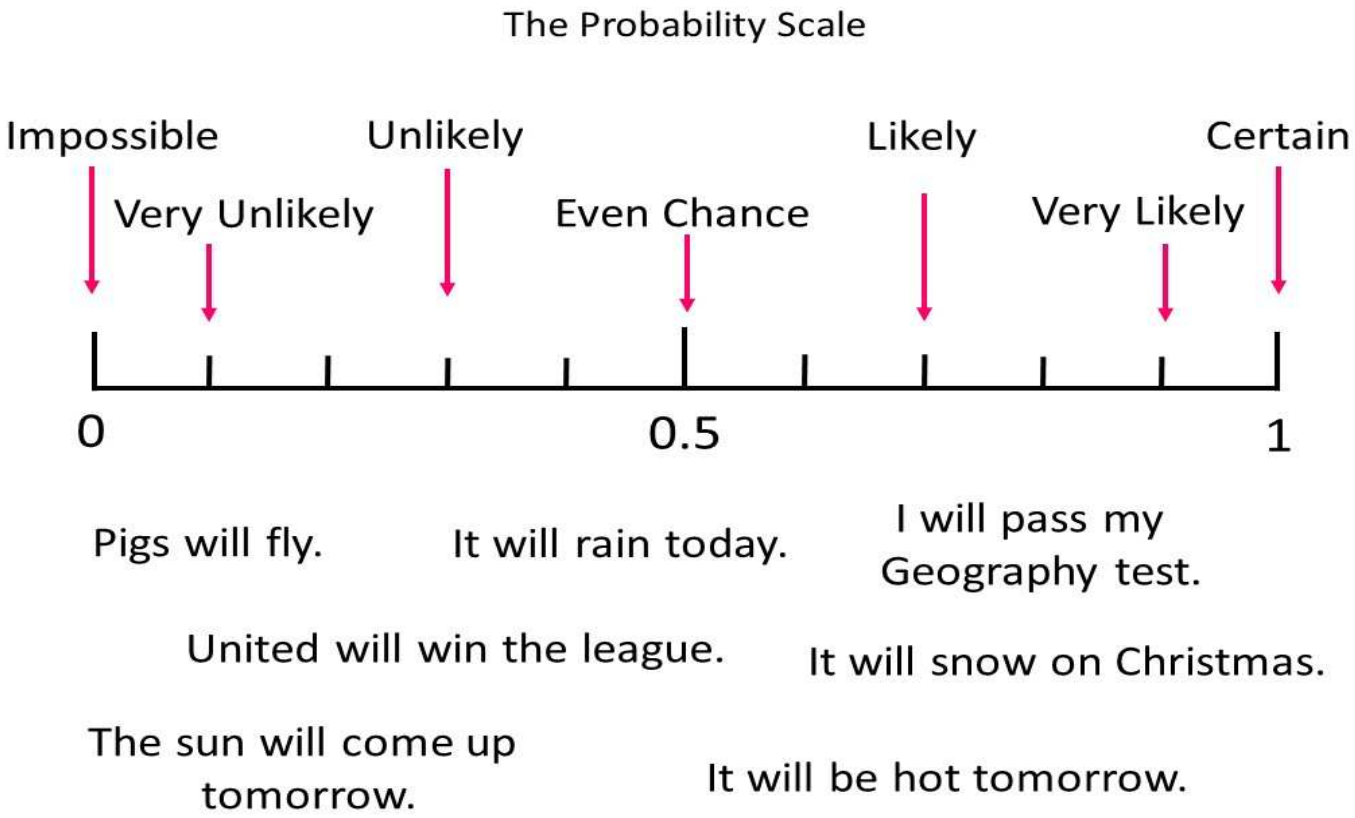
Probability & Statistics

Module 1

Probability basics

Probability scale



Basic Definitions

- **EXPERIMENT:** In the study of probability, any process of observation is referred to as experiment. The results of an observation are called the outcomes of an experiment.

- **RANDOM EXPERIMENT:** An experiment whose outcome is not known in certainty is known as random experiment.

  Example: Tossing a coin, throwing a die are random experiments.

- **TRIAL & EVENT:** An experiment which is repeated under essentially identical conditions to get some results, is known as a trial and the results are known as outcomes or events.

  Example: Tossing a coin is a trial and getting head or tail is an event.

Basic Definitions

- **EXHAUSTIVE EVENTS:** The total number of possible outcomes in any trial is known as exhaustive events.

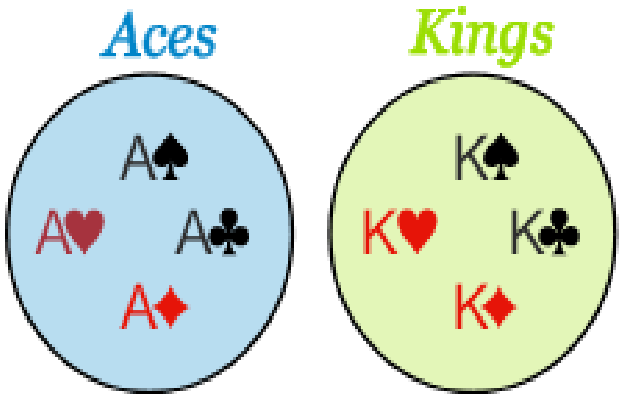
Example: In throwing a die, there are 6 exhaustive cases namely 1,2,3,4,5 and 6.

- **FAVOURABLE EVENTS:** The number of cases favourable for the occurrence of an event in a trial is known as favourable events.

Example: In a throw of a die, the number of cases favourable to get an even number is 3 as there are 3 even numbers namely 2,4 and 6 in a die.

Mutually Exclusive Events/ Disjoint Events:

- Two events are said to be mutually exclusive or incompatible, if the happening of one event prevents the happening of the other event.
- **Example:** In a tossing of a coin, the events head and tail are mutually exclusive events.



INDEPENDENT EVENTS

- Two events are said to be independent, if the happening of one event does not affect the happening of the other event.
- **Example:** In tossing 2 coins, getting head in first coin is independent of getting tail in the second coin.

Mutually Exclusive & Independent Events

$$P(A \text{ and } B) = P(A) \times P(B)$$

Independent

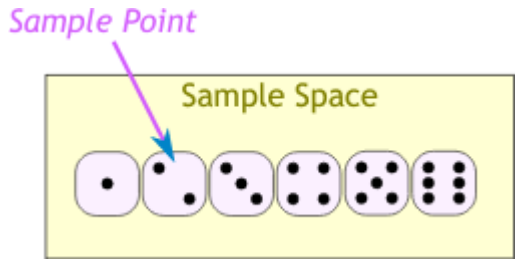
$$P(A \text{ or } B) = P(A) + P(B)$$

Mutually Exclusive

Probability

- **SAMPLE SPACE (S):** The set of all possible outcomes of a random experiment is called sample space.

  Example: The sample space in a toss of a coin is $S = \{H.T\}$.



- **PROBABILITY:** The chance of occurrence of an event in a random experiment is said to be probability.

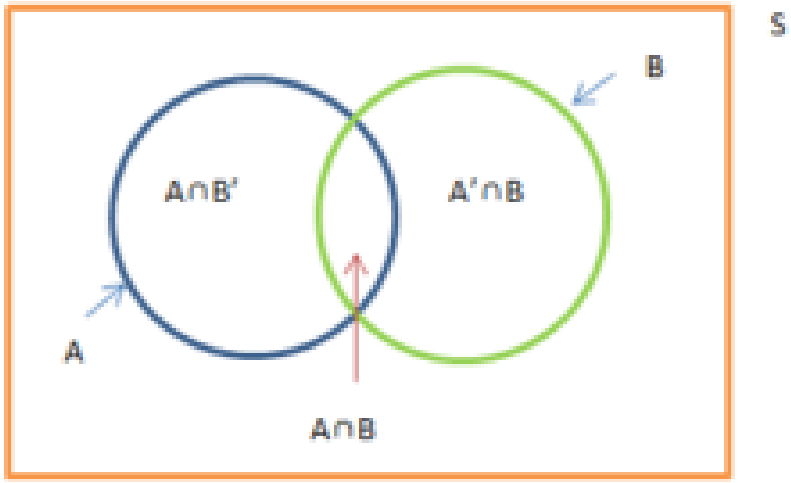
- **Definition:** Let *S* be the sample space and *A* be an event with a random experiment. Let $n(S)$ and $n(A)$ be the number of elements of *S* and *A*. Then the probability of event *A* occurring is defined by $P(A) = \frac{n(A)}{n(S)} = \frac{\text{number of cases favourable to } A}{\text{exhaustive number of cases in } S}$

Axioms Of Probability:

- Let S be the sample space and A be an event with a random experiment. Then the probability of event A , denoted by $p(A)$ is defined as a real number satisfying the following axioms.
  - $0 \leq P(A) \leq 1$
  - $P(S) = 1$
  - If $A_1, A_2, \cdots, A_n$ are mutually exclusive and exhaustive events then $P(A_1 \cup A_2 \cup \cdots \cup A_n) = P(A_1) + P(A_2) + \cdots + P(A_n)$

Addition Theorem

If A and B are any two events in a sample space S, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

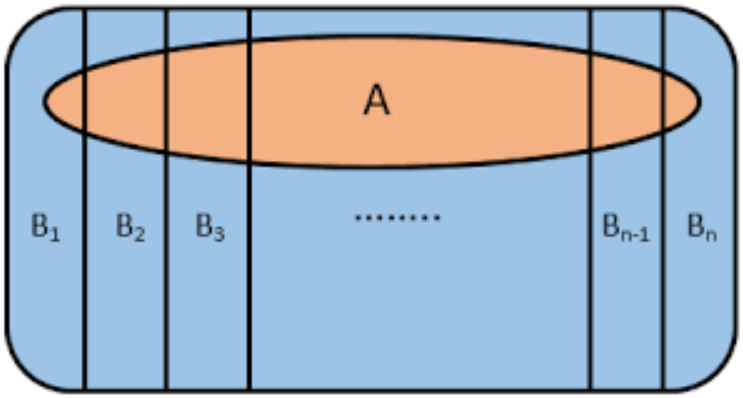


Multiplication Theorem / Conditional Probability

- The probability of an event *B* after the occurrence of an event *A*, is said to be conditional probability of *B* over A and is given by
- $P(B/A) = \frac{P(A \cap B)}{P(A)}$, provided $P(A) \neq 0$
- => $P(A \cap B) = P(B/A).P(A)$

TOTAL PROBABILTY OF AN EVENT

- If $B_1, B_2, \cdots, B_n$ are mutually exclusive and exhaustive events of a sample space *S* and A is any event in *S*, then
- $P(A) = P(A/B_1).P(B_1) + P(A/B_2).P(B_2) + \cdots + P(A/B_n).P(\ B_n).$
- The probability of event *A*, that is $P(A)$ is called the total probability of event *A*.



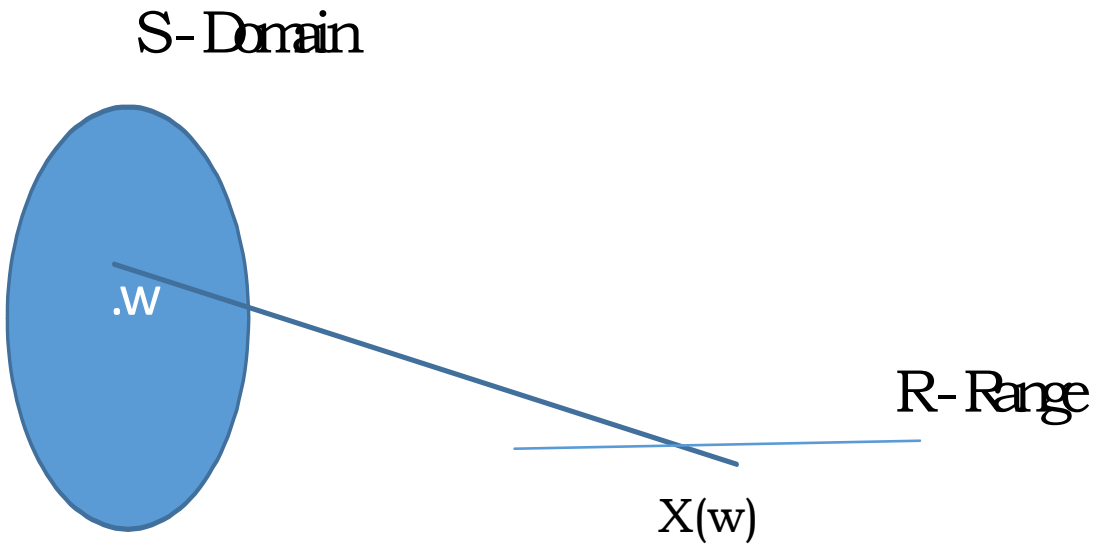
BAYE'S THEOREM

- If $A_1, A_2, \cdots, A_n$ are mutually exclusive and exhaustive events of a sample space S such that $P(A_i) > 0$ for $i = 1,2, \ldots, n$ and B is any event in S, such that $P(B) > 0$ then
- $P(A_i/B) = \frac{P(A_i).P(B/A_i)}{P(B)}$

Random Variables

Random Variables

Let S be the sample space associated with a given random experiment. A real valued function defined on S and taking values in $(-\propto, \propto)$ is called a one dimensional random variable or simply a random variable (r.v).



Types of random variables

- Discrete random variable
- Continuous random variable

Discrete Random Variable

If a random variable takes at most a countable number of values it is called a ***discrete random variable***

In other words, a real valued function defined on a discrete sample space is called a ***discrete random variable***

Probability mass function

Suppose X is a discrete random variable taking at most a countable number of values $x_1, x_2, x_3, \ldots$. Each possible outcome x_i is associated with a probability i.e. $P(X = xi) = p\,(xi) = pi$ called the probability of x_i. The number $p\,(x_i)$ must satisfy the following conditions

$$p(x_i) \geq 0$$

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

Distribution function of a random variable or Cumulative Distribution function

Let X be a discrete random variable taking countable number of values $x_1, x_2, x_3 \ldots$ with associated probabilities $p(x_i) \geq 0$. Then the distribution function of X is given by

$$F(x) = F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i), -\infty \leq x \leq \infty$$

$$P(X \leq x) = \{w \in S / X(w) \leq x\}$$

Continuous random variable

A random variable X is said to be continuous if it can take all possible values between certain limits i.e., it takes all possible values in a given interval

Probability density function

If X is continuous random variable such that

$$P(x-\frac{dx}{2} \le X \le x+\frac{dx}{2}) = f(x)dx$$

then f(x) is called the pdf of X provided f(x) satisfies the following conditions.

$$f(x) \ge 0$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Continuous distribution function

If X is a continuous random variable with pdf $f(x)$ then the function

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx \quad , -\infty < x < \infty$$

is called the distribution function or cumulative distribution function

Mean Of A Random Variable

- The mean or average or expectation of a random variable X is defined as

$$\bar{X} = E(X) = \begin{cases} \sum xp(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)dx\,, & \text{if } X \text{ is continuous} \end{cases}$$

- **RESULTS:**
  - $E(aX + bY) = aE(X) + bE(Y)$
  - $E(c) = c$, where c is a constant
  - $E(XY) = E(X).E(Y)$, if X and Y are independent

Variance Of A Random Variable

- The variance of a random variable X is defined as

$$Var(X) = E(X - E(X))^2$$

- **RESULTS:**
  - $Var(X) = E(X^2) - (E(X))^2$
  - $Var(c) = 0$, where c is a constant
  - $Var(cX) = c^2 Var(X)$
  - $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y)$, if X and Y are independent